

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 07

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Find the value of $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$.

- A $\frac{9}{4}$
- B $-\frac{4}{9}$
- C $\frac{4}{9}$
- D $-\frac{9}{4}$
- E $\frac{2}{3}$
- F $\frac{3}{2}$

- 2 A cumulative frequency graph for 80 observations passes through (20, 20), (30, 40) and (45, 60). Estimate the interquartile range.

A 40

B 10

C 20

D 25

E 15

F 30

- 3** The graph of $y = f(x)$ has a maximum point at $(3, 8)$. The graph of $y = f(x - a) + b$ has its maximum at $(1, 2)$. Find $a + b$.

- A** 8
- B** -2
- C** -4
- D** -8
- E** 4
- F** 0

- 4 The graph of $y = x^2 + bx + c$ crosses the x -axis at $(1, 0)$ and $(5, 0)$. What is the y -coordinate of the vertex?
- A** -9
- B** 5
- C** -4
- D** -3
- E** 4
- F** -5

5 Three fair coins are tossed. Find the probability of getting more heads than tails.

- A $\frac{3}{4}$
- B $\frac{1}{4}$
- C $\frac{1}{8}$
- D $\frac{5}{8}$
- E $\frac{3}{8}$
- F $\frac{1}{2}$

6 Find the exact value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.

A $\frac{\sqrt{3}}{2}$

B $\frac{3}{4}$

C $\frac{\sqrt{3}}{4}$

D 1

E 0

F $\frac{1}{2}$

7 Solve $x^{\frac{3}{2}} = 8x^{-\frac{3}{2}}$ for $x > 0$.

A $x = 2\sqrt{2}$

B $x = 16$

C $x = 8$

D $x = \sqrt{2}$

E $x = 2$

F $x = 4$

8 A triangle has sides 4, 5 and 6. Find the cosine of its largest angle.

- A** $\frac{9}{16}$
- B** $\frac{1}{8}$
- C** $\frac{5}{8}$
- D** $-\frac{1}{8}$
- E** $\frac{1}{4}$
- F** $\frac{3}{4}$

- 9 In a class of 30, 18 study French, 12 study German and 5 study both. A student is chosen at random. Given that the student studies French, what is the probability that they also study German?

- A $\frac{1}{6}$
B $\frac{5}{25}$
C $\frac{18}{5}$
D $\frac{18}{30}$
E $\frac{13}{18}$
F $\frac{5}{12}$

10 The equation $x^2 + (k - 2)x + (k + 1) = 0$ has two distinct real roots. Find the complete set of possible values of k .

A $0 < k < 8$

B $-2 < k < 4$

C $k < -2$ or $k > 4$

D $k < 0$ or $k > 4$

E $k < 0$ or $k > 8$

F all real k

11 $f(x) = x^2 - 4x$. The graph of $y = f(x + 2)$ has its minimum point at

A $(0, -4)$

B $(2, -4)$

C $(4, -4)$

D $(2, 0)$

E $(-2, -4)$

F $(0, 0)$

12 For which of the following data sets is the mean greater than the median?

A 1, 2, 3, 4, 5

B 5, 5, 5, 5, 5

C 2, 4, 6, 8, 10

D 1, 2, 3, 4, 20

E 1, 10, 11, 12, 13

- 13 Find $\frac{d^2y}{dx^2}$ when $y = \frac{1}{x^2} + 4\sqrt{x}$.
- A** $\frac{x^4}{3} - x^{-\frac{3}{2}}$
- B** $\frac{x^4}{6} + x^{-\frac{3}{2}}$
- C** $\frac{x^4}{6} - 2x^{-\frac{3}{2}}$
- D** $\frac{x^4}{6} - x^{-\frac{3}{2}}$
- E** $-\frac{2x^4}{6} - x^{-\frac{3}{2}}$
- F** $\frac{2x^4}{x^3} + 2x^{-\frac{1}{2}}$

14 The n th term of an arithmetic sequence is $u_n = 7 - 4n$. Find the sum of the first 30 terms.

A -1650

B -1530

C -1740

D -1590

E -1770

F -1860

15 How many real solutions does the equation $x^3 = 3x + 1$ have?

A 2

B 0

C 1

D 3

E infinitely many

16 How many integers x satisfy $x^2 < 5x + 24$ and $x^2 > 3x$?

A 4

B 8

C 5

D 7

E 6

F 9

- 17** The line $y = x + c$ and the curve $y = \frac{1}{x}$ have no points of intersection. Find the complete set of values of c .

A $c < -2$ or $c > 2$

B $-1 < c < 1$

C $c > 2$

D $c < 2$

E $-2 < c < 2$

F no such values

- 18** The integers from 1 to 100 inclusive are written down. What is the sum of those that are multiples of 3 or 5 (or both)?
- A** 2500
 - B** 2350
 - C** 2733
 - D** 2103
 - E** 2418
 - F** 2318

19 Which of the following inequalities has exactly the same solution set as $\frac{1}{x} > 2$?

A $x > \frac{1}{2}$

B $0 < x < \frac{1}{2}$

C $x < \frac{1}{2}$

D $x < 0$ or $x > \frac{1}{2}$

E $x > 2$

F $0 < x < 2$

20 Find the area of the finite region enclosed by the curve $y = x^2$ and the line $y = 2x + 3$.

- A** $\frac{16}{3}$
- B** $\frac{28}{3}$
- C** 12
- D** $\frac{20}{3}$
- E** $\frac{32}{3}$
- F** $\frac{64}{3}$

END OF TEST